THRIFTKART / SEMESTER III

Business Requirements Document

Product direction, business intent and MVP boundaries

Project	ThriftKart - Second-Hand Textbook Marketplace
Course	Software Engineering & Project Management
Case Study	Case Study no. 72
Audience	Project team, incubator mentor, faculty evaluator
Baseline	Supplied case study + proposed product decisions
Status	Assessment-ready working baseline

1. Executive Context

ThriftKart is a campus-focused web marketplace where senior students can list used textbooks, junior students can discover and request them, and the exchange is completed through an on-campus handover. The supplied case positions the project as a time-constrained student startup with a ₹1.5 lakh incubation grant and an exam constraint, so product scope must remain intentionally small.

Case anchor

The case specifies an average textbook spend of ₹6,500 per year, a first-semester target of 500 books listed and 150 transactions, and a team of four students available for 3 hours per weekday except exam weeks.

2. Business Problem

Students pay for books that may be used for only one semester, while seniors often keep books with little reuse value. A campus marketplace can connect these two groups, shorten the path from listing to handover and make the process more trustworthy than informal group chats or scattered listings.

3. Business Objectives

ID	Objective	Expected outcome
BR-01	Affordable access	Enable students to buy used textbooks at lower cost and let seniors recover value from books that are no longer needed.
BR-02	Campus-only exchange	Keep handover on campus so buyer and seller can coordinate a simple, familiar exchange without delivery complexity.
BR-03	Trust before scale	Require moderation before publication and retain evidence of moderation decisions.
BR-04	Semester traction	Use the case-given first-semester targets of 500 books listed and 150 transactions as the primary business outcome.
BR-05	Finishable product	Limit the MVP to the workflows the four-student team can realistically deliver before the exam constraint.
BR-06	Evidence-led delivery	Maintain estimates, requirements traceability, systematic tests, defect metrics and risk controls as part of project completion.

4. Stakeholders & Needs

Stakeholder	Primary need
Buyer / junior student	Find suitable books quickly; compare price/condition; message seller; coordinate handover; know transaction state.
Seller / senior student	Create a listing quickly; provide accurate condition; receive buyer requests; confirm handover; close the sale.
Admin moderator	Review listings; reject unsafe/incomplete posts; handle reports; leave an auditable reason.
Incubator mentor	See a coherent plan, realistic scope, measurable requirements and evidence of progress.
Project team	Deliver a finishable MVP before exam weeks and demonstrate software engineering artefacts.

5. Current-to-Future Process

Stage	Business definition

Stage	Business definition
Current pain	Books are underused after one semester; discovery and negotiation are fragmented.
Future flow	Register → browse/search → view listing → chat or request → confirm payment status → campus handover → complete transaction.
Control point	Every seller listing passes through Pending Moderation before public discovery.
Outcome	A traceable exchange record is created for each completed sale.

6. MVP Scope

MVP boundary

Core: account access, listing creation, price/condition validation, moderation, search/filter, listing details, purchase request, transaction status and campus handover. Chat is lightweight. A full online payment gateway is intentionally deferred; the MVP records payment method/status and supports external settlement or cash on handover.

7. MoSCoW Priorities

Priority	Items	Reason
Must	FR-01, FR-03, FR-04, FR-05, FR-06, FR-07, FR-09, FR-10, FR-11	Core marketplace and sale path
Should	FR-02, FR-08, FR-12	Important trust/usability enhancements
Could	Saved searches, advanced recommendations, delivery, automated dispute handling	Not required for MVP
Won't (MVP)	Full bank/UPI gateway integration, logistics/delivery network, complex recommendation engine	Explicitly outside the time-constrained MVP

8. Business Rules

Rule	Name	Definition
BRULE-01	University email required	Only verified university-domain accounts can use buyer/seller flows.
BRULE-02	Moderation gate	A listing cannot be public unless its moderation decision is Approved.
BRULE-03	Price validation	Proposed product rule: price must be a positive whole-number amount up to ₹20,000.
BRULE-04	Condition validation	Condition rating must be an integer from 1 (poor) to 5 (excellent).
BRULE-05	Single active reservation	A listing may have at most one active reservation at a time.
BRULE-06	Campus handover	Completed transactions require a handover point and a completion confirmation.
BRULE-07	Auditability	Every moderation action stores moderator, decision, reason and timestamp.

9. Success Measures

500	150	100%	9–10
books listed, first semester	transactions, first semester	listings moderated before publish	exam-week constraint

The two numerical traction targets above are directly supplied by the case. The moderation metric is a proposed control derived from the trust-and-safety objective; the exam-week notation reflects the case statement that team availability is zero in weeks 9 and 10.

10. Acceptance Conditions

- Buyer can locate a published book using search/filter and reach a listing detail page.
- Seller can submit a book for moderation only when required fields, price and condition are valid.
- Admin can approve/reject a listing and record a reason; rejected listings do not appear in the public marketplace.
- A purchase request cannot create conflicting active reservations for the same listing.
- A completed transaction contains buyer, seller, listing, agreed price and handover information.
- Project artefacts remain traceable from business requirement to system requirement to test case.

11. Assumptions & Constraints

Type	Statement
A-01	The university provides a verifiable student/university email domain.
A-02	Campus handover is available and accepted as the standard fulfillment model.
A-03	The MVP may use an external payment method or cash on handover instead of implementing a full gateway.
C-01	Team capacity is 4 students × 3 hours/day on weekdays, with zero hours in weeks 9 and 10.
C-02	The project must remain small enough to complete within the exam constraint; the case explicitly asks for an MVP.

12. Stakeholder Approval

Role	Name	Signature / date
Project sponsor / mentor		
Faculty evaluator		
Student project team		